Change Detection Based on Supervised Contrastive Learning for High-Resolution Remote Sensing Imagery

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***Abstract*— Change detection (CD) is a challenging task on high-resolution bitemporal remote sensing images. Many recent studies of CD have focused on designing fully convolutional Siamese network architectures. However, most of these methods initialize their encoders by random values or an ImageNet pretrained model, without any prior for the CD task, thus limiting the performance of the CD model. In this article, the novel supervised contrastive pretraining and fine-tuning CD (SCPFCD) framework, which is made up of two cascaded stages, is presented to train a CD network based on a pretrained encoder. In the first supervised contrastive pretraining stage, the encoder of the Siamese network is asked to solve a joint pretext task introduced by the proposed CDContrast pretraining method on labeled CD data. The proposed CDContrast pretraining method includes land contrastive learning (LCL), which is based on supervised contrastive learning, and proxy CD learning. The LCL focuses on learning the spatial relationships among the land cover from bitemporal images by solving a land contrast task, while the proxy CD learning performs a proxy CD task on the top of the upsampling projector to avoid local optima for the LCL and learn features for the CD. Then, in the second fine-tuning stage, the whole Siamese network initialized with the pretrained encoder is fine-tuned to perform the CD task in an end-to-end manner. The proposed SCPFCD framework was verified with three CD datasets of high-resolution remote sensing images. The extensive experimental results consistently show that the proposed framework can effectively improve the ability to extract change information for Siamese networks.**

***Index Terms*— Change detection (CD), deep learning, multitemporal image analysis, pretraining, supervised contrastive learning (CL).**

I. Introduction

THE aim of change detection (CD) is to identify differences in the state of the objects of interest by observing the scene at different times [1]. CD with high-resolution bitemporal remote sensing images has been widely used in various applications, such as city planning, land-cover monitoring, and disaster emergency response, which has prompted the development of various CD methods [2], [3], [4], [5], [6].

In a general CD method, the change/no change information or features are first extracted from two registered bitemporal remote sensing images and are then used to obtain the change map. Instead of using handcrafted operators, recent CD methods have used deep learning technology, including fully convolutional neural networks (FCNs) [3], [4], [5], [7], [8], [9], [10], [11] and transformers [12], [13], to extract powerful features for the CD task. In these methods, the CD task is regarded as a binary classification task, i.e., the FCN and transformer are employed to identify the change or no change for each bitemporal pixel pair. Although transformers have shown superiority in the CD field, applying the transform to pixel classification task still faces many challenges, such as lack of inductive biases [14]. Therefore, many researchers still work on FCN-based CD methods, especially its essential branch, i.e., the Siamese network, which is the current research hotspot in CD field [5], [7], [8], [9], [10], [11].

Most of the existing Siamese networks adopt random values [5], [7], [8], [9] or an ImageNet pretrained model [10], [11] to initialize the parameters of their encoders. Random initialization is usually used for a Siamese network with a lightweight encoder, which can avoid overfitting, but random initialization lacks any prior for the CD, limiting the performance of the Siamese network. Although an ImageNet pretrained model can provide a powerful parameter initialization for a Siamese network, the CD performance will be limited by the domain gap between the natural imagery and the remote sensing imagery [15]. Furthermore, due to the high cost of high-resolution images and human annotation, the size of a labeled CD dataset is usually limited [4], [16], which makes the parameter initialization approach for a Siamese network become more important [17].

Recently, some researchers have used unsupervised contrastive learning (CL) to pretrain an encoder on unlabeled low- and medium-resolution multitemporal remote sensing images to obtain a good parameter initialization for the downstream CD task [15], [18], [19], [20]. Without any human annotation, all these methods regard two remote sensing image patches at different locations as negative sample pairs [15], [18], [19], [20] and regard multitemporal image patches at the same location as positive sample pairs [15], [18], [20] for unsupervised CL. However, the land covers of different locations can belong to the same class, while the land covers at the same location may change a lot in bitemporal images, which introduces noise to the CL, and, thus, limits the efficiency of pretraining, especially on small amounts of data [18], [19], [20], [21]. Although Mañas et al. [15] carefully collected more than 1M multitemporal low- and medium-resolution images to guarantee the effectiveness of unsupervised CL,

Manuscript received 1 August 2022; revised 5 November 2022 and 29 November 2022; accepted 29 November 2022. Date of publication 13 January 2023; date of current version 24 January 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 42071350 and in part by the LIESMARS Special Research Funding. *(Corresponding author: Yanfei Zhong.)*

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Digital Object Identifier 10.1109/TGRS.2023.3236664

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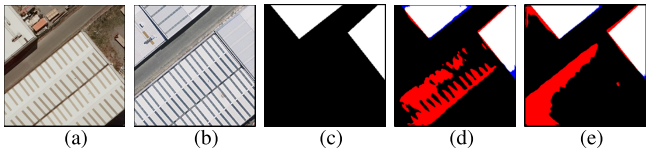


Fig. 1. Example from the WHU building dataset [22]. (d) and (e) Change maps obtained by the IFN and SNUNet (□ TPs, ■ FPs, ■ FNs, and ■ TNs). (a) T_1. (b) T_2. (c) GT. (d) IFN. (e) SNUNet.

it is expensive to collect multitemporal high-resolution images at such a scale [16], and it would use a lot of computing resources to perform CL on the data. Although the pretrained models of Mañas et al. [15] are available and can be used to initialize a Siamese network for high-resolution images, the domain gap between the different-resolution images can limit the CD performance. This is because CD on high-resolution images is faced with different challenges from CD on low- and medium-resolution images, including the large reflectance variability among the different land-cover classes and the different imaging characteristics [2]. Fig. 1 shows a representative situation of CD on bitemporal high-resolution images, achieved by two recent advanced Siamese networks, namely, the image fusion network (IFN) [10] and SNUNet [9]. The big radiation differences and complex textures on the roofs of the high-resolution remote sensing images cause many false alarms. More recently, Chen et al. [6] proposed the SaDL method, which is based on CL using labels and effectively pretrains encoder for building CD on high-resolution images. However, SaDL needs to collect additional single-temporal images with annotated all the concerned land covers, the high cost of which limits SaDL to pretrain for CD involving more land cover types.

In this article, the supervised contrastive pretraining and fine-tuning CD (SCPFCD) framework based on supervised CL is proposed to train an FCN-based Siamese network on the basis of an encoder with good parameter initialization for high-resolution remote sensing image CD. The supervised CL technique, which effectively uses *labels* to construct sample pairs for contrasting, has succeeded in pretraining a powerful encoder for the downstream tasks [23], [24]. In the proposed SCPFCD framework, supervised CL is used to pretrain an encoder of a Siamese network for the CD with labeled bitemporal high-resolution remote sensing images. Our contributions can be summarized as follows.

1) The SCPFCD framework includes two cascaded stages, i.e., supervised contrastive pretraining and fine-tuning, and is proposed to train a Siamese network based on an encoder with good parameter initialization for the CD task. To achieve this, in the first supervised contrastive pre-training stage, the proposed CDContrast pretraining method, which is asked to solve a joint pretext task of land CL (LCL) based on supervised CL and proxy CD learning, aims to pretrain the encoder on the labeled CD data for the downstream CD task. In the second fine-tuning stage, the pretrained encoder is fine-tuned to perform the CD task with a CD decoder. In this way, the CD ability of a Siamese network can be improved, based on good parameter initialization.
2) The LCL is proposed to learn the spatial relationships among the land cover for the encoder in the supervised contrastive pretraining. In the LCL, the land contrast task regards bitemporal unchanged pixel pairs as positive samples and asks the encoder to enhance the aggregation of the positive samples in the upsampled projection space, where the information loss caused by the downsampled labels can be reduced. By solving the land contrast task, the encoder of the Siamese network is optimized and obtains a visual representation of the land cover for the downstream CD task on bitemporal images.
3) Proxy CD learning is proposed to prevent the LCL from falling into a local optimal solution and learn a feature representation involving the CD in the supervised contrastive pretraining. In proxy CD learning, the encoder is asked to solve a proxy CD task of identifying change and no change in a CD guider, using the bitemporal upsampled projections. The proxy CD learning works together with the LCL, so as to guarantee the feature quality for the downstream CD task.

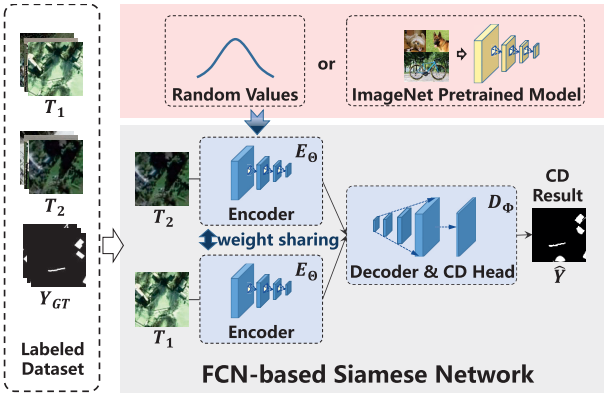


Fig. 2. Illustration of a typical fully convolutional Siamese network.

The experimental results obtained on three public CD datasets of high-resolution remote sensing images consistently show that the SCPFCD framework can achieve a significantly better CD performance than the models without supervised contrastive pretraining.

The rest of this article is organized as follows. Section II introduces the recent fully supervised Siamese networks for CD, the CD methods that use the CL technique, and the recent pretraining methods based on the supervised CL technique. Section III describes the proposed SCPFCD framework and the CDContrast pretraining method. In Section IV, the quantitative experimental results obtained on the three datasets are reported. Meanwhile, feature analysis is conducted with land-cover labels to explain why the proposed framework works so well. Section V describes the ablation study conducted to analyze the effect of each module of the proposed framework, and also the space and time complexities. Finally, Section VI gives our conclusions and the potential future work.

II. Related Work

A. Fully Supervised Siamese Networks for Change Detection

As shown in Fig. 2, an FCN-based Siamese network can be divided into three parts: Siamese encoder, decoder, and CD head [7]. The Siamese network first passes the two temporal images through two weight-sharing Siamese encoders, extracting features for the two bitemporal images. The encoder

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features of the two temporal images are then input into the decoder to recover their spatial size to the size of the input bitemporal images and extract the decoder feature that contains change information. The CD head is a linear binary classifier, mapping the decoder feature to the change map.

Daudt et al. [7] proposed the first fully convolutional Siamese network—FC-Siam-conc—for CD, which is based on UNet [25]. By the use of a simple CD decoder for extracting change information, Daudt et al. [7] also proposed the FC-Siam-diff (FC-DIFF) model. To fuse multilevel features, Zhang et al. [10] proposed IFN, where the attention mechanism is introduced to the decoder of a Siamese network. Similarly, Song et al. [8] proposed SUACDNet with residual structure and three types of attention modules to capture edge detail information for CD. Lee et al. [11] introduced a new change attention map-based content loss to a Siamese network to perform CD for complex urban areas. Subsequently, Zhang et al. [5] introduced superpixel sampling networks into a UNet-based Siamese network for the CD task to overcome the latent noise problem brought by the feature maps of FCNs. Fang et al. [9] proposed SNUNet with an ensemble channel attention module to handle the localization information loss in a Siamese network for CD.

All in all, to improve the CD performance, these methods focus on designing an effective architecture for the Siamese network, and they adopt the common parameter initialization approaches, namely, random values or an ImageNet pretrained model. Due to the lack of any prior for the CD task, the performance of these methods is limited by the parameter initialization method, especially when labeled data are not abundant [17]. In this article, the proposed CDContrast pretraining method pretrains the encoder of the Siamese network, which is then fine-tuned on the labeled CD data. Hence, the proposed SCPFCD framework is able to improve the CD ability of a Siamese network and is compatible with many of the existing Siamese networks.

B. Change Detection Based on Contrastive Learning

Unsupervised CL has been successful in pretraining a powerful encoder for the downstream task in natural images [26], [27], [28], [29], which has motivated researchers to use the CL technique to pretrain an encoder for CD on unlabeled multitemporal low- and medium-resolution Sentinel-2 remote sensing images [15], [18], [19], [20]. All these methods consider bitemporal images at different locations as negative sample pairs, while Mañas et al. [15], Leenstra et al. [18], and Saha et al. [20] considered two images at the same location as a multiview positive sample pair. These methods work based on the strong assumption that the land covers of the same location do not change in two temporal images, and the land covers of different locations differ. Therefore, the performance of these unsupervised CL methods may be limited by the noise when this assumption fails, especially on small amounts of data [18], [19], [20]. Since massive and diverse samples are beneficial for the feature quality of the pretrained encoder [21], [26], [27], Mañas et al. [15] collected image patches from more than 200k locations around the world and used 1M multitemporal free Sentinel-2 images of 10k cities to pretrain the encoder. However, it is too expensive to collect such a volume of high-resolution multitemporal remote sensing images [16], and huge amounts of computing resources would be required to pretrain an encoder on unlabeled data of such scale.

More similar to our work, Chen et al. [6] proposed the SaDL method based on CL, which uses labels and image augmentation to obtain multiview positives to pretrain encoders for building CD on high-resolution images and achieves significant CD improvements compared with other pretraining methods. However, SaDL has to collect additional single-temporal images with building labeled by human experts for pretraining, which will make it cost too much to generalize to CD involving more land cover categories.

Inspired by the afore-mentioned existing methods [15], [18], [20], the method proposed in this article also uses bitemporal images as multiview positive sample pairs but using CD labels to construct positive pixel pairs without collecting additional data for pretraining. This allows the proposed SCPFCD framework to effectively pretrain an encoder on limited-size CD data without noise in the LCL based on supervised CL. Moreover, the LCL uses an upsampling projector to reduce the information loss caused by downsampling the CD labels. Meanwhile, the proxy CD learning is used to avoid the dilemma of the CL, ensuring the optimization direction for the downstream CD task. In this way, the feature quality of the pretrained encoder is guaranteed for the CD task.

C. Pretraining Based on Supervised Contrastive Learning

In the computer vision field, supervised CL uses both an image with different data augmentation as different views and the labels to construct sample pairs for representation learning, so it avoids the noise introduced by separating the features of the same class in the projection space [27], [30]. Therefore, supervised CL can effectively pretrain an encoder on a labeled dataset of limited size in many scenarios. Inspired by SimCLR [27], Khosla et al. [30] first proposed a supervised CL to pretrain an encoder for the downstream task. Cui et al. [31] proposed PaCo based on supervised CL to pretrain an encoder for long-tailed recognition. Cao et al. [24] proposed a supervised CL method to pretrain an encoder for mammographic triage screening. More similar to our work, Zhao et al. [23] proposed a pixel-level supervised CL method, where an encoder is pretrained and then fine-tuned to perform the semantic segmentation task, on the same labeled dataset.

There are three major differences between the proposed CDContrast pretraining method and the supervised CL-based pretraining methods.

1) Instead of using data augmentation to obtain different views for one sample, the proposed LCL in the CDContrast pretraining method regards bitemporal pixel pairs at the same location as different views and uses the CD labels to construct pixel-level sample pairs to effectively learn the spatial feature of the land cover for bitemporal images.
2) Compared with objects in natural images, the land cover in high-resolution remote sensing images contains rich details. This limits the pretraining due to the loss of

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information occurring in the CD label downsampling. To alleviate this problem, an upsampling projector, which includes a convolution-based multilayer perceptron (MLP) and upsampling, is proposed in the LCL.

3) The downstream task of the afore-mentioned CL methods is usually concerned with only one image, while CD involves analyzing bitemporal images. Furthermore, the features for the land cover and for locating changes are inconsistent [32]. Therefore, pretraining only by the CL technique may lead to local optima for the feature learning for the spatial relationships of the land cover and can make the encoder fail to obtain the CD ability. To this end, proxy CD learning is proposed to adjust the optimization direction of the supervised CL for the downstream CD task.

III. Proposed SCPFCD Framework

As shown in Fig. 3, the SCPFCD framework is divided into two main stages. Under the supervision of the CD labels, the first supervised contrastive pretraining stage aims to pretrain the encoder for the downstream CD task based on the proposed CDContrast pretraining method. In the second fine-tuning stage, after loading the pretrained encoder in the first stage, the fully convolutional Siamese network is trained by the same CD labels, so as to obtain the ability of CD.

A. Overall of the SCPFCD Framework

The two stages of the SCPFCD framework are trained by exactly the same bitemporal remote sensing images T_1 and T_2 and the CD labels Y_{GT} from the training set. The different optimization targets of the two stages lead to the different feature representations that the encoder is able to learn.

1) Supervised Contrastive Pretraining Phase: An encoder E_θ, where θ represents its learnable parameters, is pretrained to solve a pretext task. Supervised by the specific loss L_{pt} of a pretext task of "contrasting," the parameters $\theta^{(i)}$ of the encoder E_θ for the ith iteration are updated as follows:

$$\theta^{(i+1)} = \theta^{(i)} - \eta \frac{\partial L_{\mathrm{pt}}}{\partial \theta^{(i)}} \tag{1}$$

where η is the learning rate. Finally, the first pretraining stage ends with a well-pretrained encoder E_Θ. As shown in the bottom gray area of Fig. 3, the deep blue encoder denotes the pretrained encoder.

2) Fine-Tuning Phase: Equipped with the pretrained encoder E_Θ, a Siamese network experiences end-to-end training for the CD task. Features are extracted in order by the pretrained encoder and the decoder from the given bitemporal images T_1 and T_2. Finally, the features are classified as "change" or "no change," so as to predict the change map $\hat{Y}$. This process is shown in the bottom gray area of Fig. 3 and can be described, as shown in the following equation:

$$\hat{Y} = D_\Phi(E_\Theta(T_1), E_\Theta(T_2)) \tag{2}$$

where D denotes the decoder and the CD head, the learnable parameters of which are denoted by Φ. With parameters Θ and Φ, the Siamese network is updated as follows:

$$\begin{aligned} \Theta^{(i+1)} &= \Theta^{(i)} - \eta \frac{\partial L_{\mathrm{CD}}\left(\hat{Y}, Y_{\mathrm{GT}}\right)}{\partial \Theta^{(i)}} \\ \Phi^{(i+1)} &= \Phi^{(i)} - \eta \frac{\partial L_{\mathrm{CD}}\left(\hat{Y}, Y_{\mathrm{GT}}\right)}{\partial \Phi^{(i)}} \end{aligned} \tag{3}$$

where L_{CD} represents the loss of the CD task and Y_{GT} is the CD ground truth.

B. CDContrast Pretraining Method

The CDContrast pretraining method is proposed to pretrain an encoder E_θ for the downstream CD task. The CDContrast pretraining method is defined by a joint pretext task, which involves a land contrast task of LCL and a proxy CD task of proxy CD learning. Correspondingly, the pretext task loss L_{pt} of the CDContrast pretraining method consists of the LCL loss L_{LCL} and the proxy CD loss L_{PCD}

$$L_{\mathrm{pt}}(T_1, T_2, Y_{\mathrm{GT}}) = L_{\mathrm{LCL}}(T_1, T_2, Y_{\mathrm{GT}}) + \lambda L_{\mathrm{PCD}}(T_1, T_2, Y_{\mathrm{GT}}) \tag{4}$$

where λ denotes the weight to balance the two losses. In the LCL, the unchanged bitemporal land-cover projections mapped from the encoder features are regarded as multiview projections to be contrasted to learn the spatial relationships among the land cover in one image when pretraining the encoder. Meanwhile, using the bitemporal land-cover projections, the proxy CD learning makes the encoder learn to identify change and no change, which also asks the land-cover features of the same class to be compact. Because the optimization directions of the two losses are partially consistent, λ is set to be 1 in the pretext task loss L_{pt}.

To be more specific, the CDContrast pretraining method includes the LCL, an upsampling projector (which is an important part of the LCL), and proxy CD learning, which are further described in Sections III-C–III-E.

C. Land Contrastive Learning

LCL is proposed to help the encoder E_θ learn the spatial relationships among the land cover. In the LCL, a land contrast pretext task is solved, where two bitemporal images are required to be contrasted. In particular, inspired by Grill et al. [28] and Xie et al. [29], the unchanged regions in the two images are regarded as multiviews of the same class land covers, namely, the positives, the similarity of which is maximized in the projection space, to learn feature representations. To perform the land contrast task, as shown in the pretraining stage of Fig. 3, the LCL includes an online network (blue) and a target network (green), which are bitemporal input images regarded as multiviews. The online network is then used to predict the target network representation. The online network is updated by gradient descent, while the target network is asymmetric to the online network and updated by momentum without gradient, which allows the pretraining to be convergent without using negatives [28], [33].

Specifically, the online network is defined by a set of learnable parameters θ and consists of three submodules, i.e., an encoder E_θ to be pretrained, an upsampling projector U_θ,

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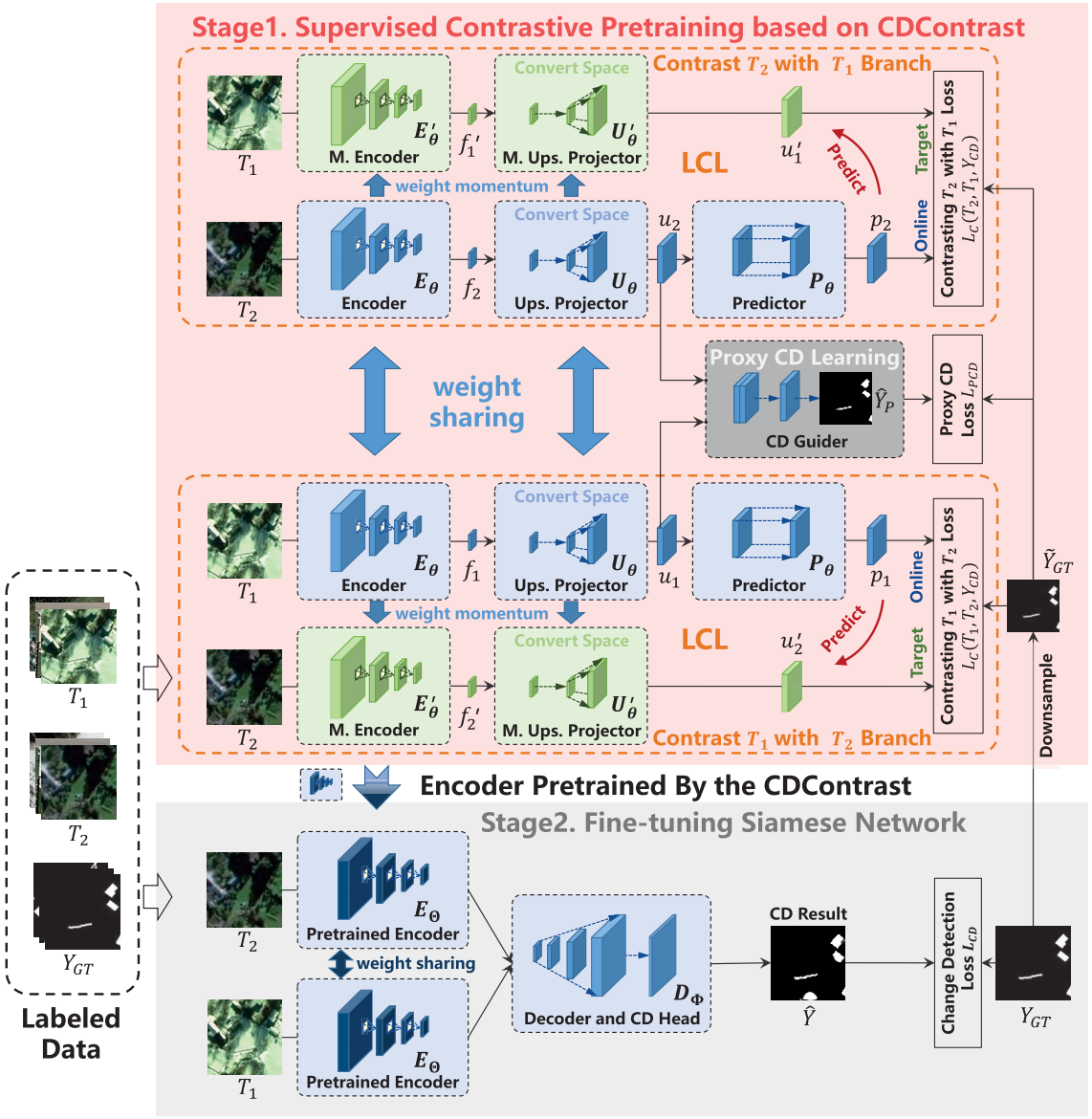


Fig. 3. Illustration of the proposed SCPFCD framework and the CDContrast pretraining method based on supervised CL. T_1 and T_2 denote the bitemporal remote sensing images, while Y_{GT} denotes the CD ground truth. In the first pretraining stage (pink area), an encoder E_θ is pretrained by CD labels. "M." denotes momentum, which means that the parameters of the module are momentum versions of the other related modules. "Ups." means upsampling. Subsequently, in the fine-tuning stage (gray area), the pretrained encoder E_Θ combined with a decoder is fine-tuned to learn the CD task using the same CD labels.

and a predictor P_θ. The upsampling projector U_θ is a simple network to improve the quality of the features extracted by E_θ. The predictor P_θ is a module used to achieve asymmetry between the online network and the target network. Here, following Xie et al. [29], the predictor P_θ is implemented by a pixel propagation module to enhance the features for the pixel-level task. As in (1), the parameters θ are updated by stochastic gradient descent (SGD) with the loss L_{LCL} of the LCL.

As for the target network, it consists of a momentum encoder $E_{\theta'}$ and a momentum upsampling projector $E_{\theta'}$, where the momentum operation is used to improve the performance of the pretraining [33]. Here, the parameters θ' are the exponential moving average (EMA) of parameters θ of the online network, namely,

$$\theta' = \mu\theta' + (1-\mu)\theta \tag{5}$$

where μ is the EMA coefficient. Here, following the setting of [29], μ is computed as

$$\mu = 1 - 0.99(\cos(\pi k/K) + 1) \tag{6}$$

where k denotes the current training iteration and K denotes the total number of training iterations.

The LCL pretrains the encoder E_θ by enforcing the positive output of the online network to predict the positive output of the target network, which refers to "contrasting." Concretely, when two samples from one positive sample pair are input into the online network and the target network, respectively, the outputs of the two networks are then forced to be gathered together. Owing to the asymmetric networks and the EMA mechanism, the contrastive behavior of the LCL, which only uses positive samples, can effectively aggregate features of the same class, without causing mode collapse [28], [29], [34]. To learn features for the land covers of multitemporal

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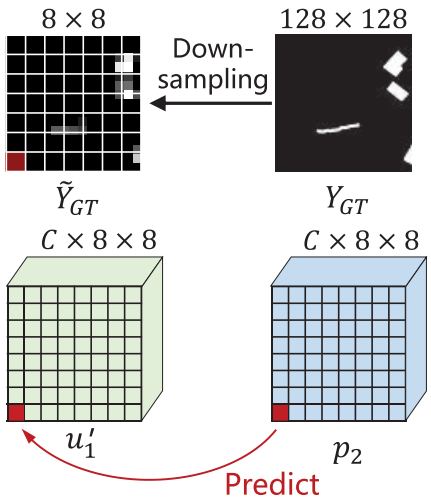


Fig. 4. Illustration of contrasting T_2 with T_1. Suppose that the output size of both the online and target networks is $C \times 8 \times 8$. For example, the highlighted red point of the downsampled ground truth $\tilde{Y}_{\mathrm{GT}}$ at the coordinates of $(0, 7)$ is an unchanged position. For the output prediction p_2 of the online network and the output upsampled projection u_1' of the target network, $p_2(0, 7)$ and $u_1'(0, 7)$, both of which are C-dimensional vectors, are encouraged to be gathered together.

images, two unchanged pixels sampled from the same position of bitemporal images T_1 and T_2 are considered as a positive sample pair for the LCL. This is because the CD labels provide the positions of the unchanged pixel pairs, which can be regarded as belonging to the same land-cover class.

Because of the asymmetric architecture of the online and target networks, the LCL includes two symmetric parts: contrasting T_1 with T_2 and contrasting T_2 with T_1, which are shown in Fig. 3. We take contrasting T_2 with T_1 as an example, which is implemented by forcing the output p_2 of the online network to predict the output u_1' of the target network, as shown in Figs. 3 and 4. In this context, with the input T_2 image, the online network successively outputs feature map f_2, upsampled projection u_2, and prediction p_2, which can be described as

$$f_2 = E_\theta(T_2), \quad u_2 = U_\theta(f_2), \quad p_2 = P_\theta(u_2) \tag{7}$$

where feature map f_2 is of size $c \times h \times w$, corresponding to the number of channels, height, and width. u_2 and p_2 are of size $C \times H \times W$. Similarly, with the input T_1 image, the target network successively outputs feature map f_1' and upsampled projection u_1', which can be described as

$$f_1' = E_{\theta'}(T_1), \quad u_1' = U_{\theta'}(f_1) \tag{8}$$

where f_1' and u_1' are of the same size as f_2 and u_2, respectively.

In this sense, as indicated in Fig. 4, the loss of contrasting T_2 with T_1 can be defined as cosine distance [28], [29]

$$L_C(T_2, T_1, Y_{\mathrm{GT}}) = \frac{1}{n_{\mathrm{uc}}} \sum_{i,j} 1 - \cos\langle p_2(i, j), u_1'(i, j)\rangle$$
$$\tilde{Y}_{\mathrm{GT}}(i, j) = 0, \quad 0 \le i \le W, \quad 0 \le j \le H \tag{9}$$

where $\cos < v_1, v_2 >$ denotes the cosine of two nonzero vectors v_1 and v_2. $\tilde{Y}_{\mathrm{GT}}$ of size $H \times W$ is a downsampled version of the CD ground truth Y_{GT} to adapt to the spatial size of u_2 and u_1' for the LCL. Here, the spatial size of u_2 and u_1' is determined by the input image size, the downsampling of the encoder E_θ, and the upsampling of the upsampling projector U_θ. $\tilde{Y}_{\mathrm{GT}}(i, j) = 0$ means that the pixels at position (i, j) do not change, i.e., they are a positive pixel pair. n_{uc} is the number of unchanged positions in $\tilde{Y}_{\mathrm{GT}}$. $p_2(i, j)$ and $u_1'(i, j)$ denote the prediction vector of size C and the upsampled projection of size C, respectively. As in (9), the loss of contrasting T_1 with T_2 can be represented as

$$L_C(T_1, T_2, Y_{\mathrm{GT}}) = \frac{1}{n_{\mathrm{uc}}} \sum_{i,j} 1 - \cos\langle p_1(i, j), u_2'(i, j)\rangle$$
$$\tilde{Y}_{\mathrm{GT}}(i, j) = 0, \quad 0 \le i \le W, \quad 0 \le j \le H. \tag{10}$$

Combining the $L_S(T_1, T_2, Y_{\mathrm{GT}})$ loss of contrasting T_1 with T_2 and the $L_S(T_2, T_1, Y_{\mathrm{GT}})$ loss of contrasting T_2 with T_1, the loss of the LCL for the land contrast task is calculated as

$$L_{\mathrm{LCL}}(T_1, T_2, Y_{\mathrm{GT}}) = L_C(T_1, T_2, Y_{\mathrm{GT}}) + L_C(T_2, T_1, Y_{\mathrm{GT}}). \tag{11}$$

In the LCL, the proposed upsampling projector plays an important part, especially for remote sensing images, which is explained in Section III-D.

D. Upsampling Projector of LCL

In the LCL, the upsampling projector is proposed to alleviate the loss of the complex and detailed information of the downsampled CD labels. The high-level but low-resolution features output by the top of an encoder with several downsampling operations are usually used to obtain projections of the same size for CL [23], [26], [27], [28], [29]. In this context, for pixel-level CL, labels are also downsampled to the size of the projections [23], [29]. However, compared with objects of natural images, the land covers of remote sensing images often appear to be small and fragmentary [35], [36], leading to small and fragmentary CD labels. Therefore, the pretraining for high-resolution remote sensing images can probably be more sensitive to the information loss of downsampled labels than that for natural images. To alleviate this problem, an intuitive motivation is to reduce the downsampling multiples of labels to reduce the information loss of downsampling the CD labels.

As shown in Fig. 5, the proposed upsampling projector comprises a convolution-based MLP followed by an upsampling unit. The convolution-based MLP is a "conv-ReLU-conv" architecture to convert the input encoder feature map f to projections [23], [29], the spatial size of which is increased by the upsampling unit to obtain the upsampled projection u. If we suppose that the size of feature f is $C \times 4 \times 4$, then the spatial sizes of the upsampled projection u_1', prediction p_2, and downsampled CD ground truth $\tilde{Y}_{\mathrm{GT}}$ are all enlarged to 8×8 with the upsampling projector of scale factor 2. The supervised CL is then performed in the projection space, corresponding to Fig. 4.

However, while the LCL regards the unchanged bitemporal pixels as multiview positive samples to learn the spatial relationship of the land cover, the features for the land cover and the changes are not fully consistent [32]. This may result in a risk of the encoder falling into a local optimal solution, where the encoder fails to learn features involving change information in the supervised contrastive pretraining stage. We describe how this problem is solved in Section III-E.

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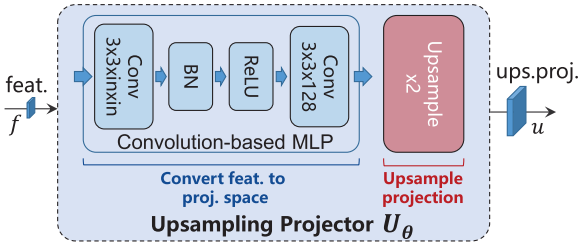


Fig. 5. Illustration of the upsampling projector. It converts the input feature map f to an upsampled projection u by a series of nonlinear operations and increases the spatial size of the projection by an upsampling unit.

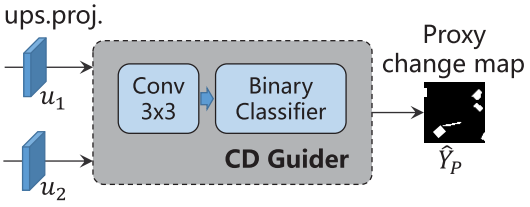


Fig. 6. CD guider for the proxy CD learning. It conducts the CD task on top of the upsampled projection, which guarantees that the features are beneficial for the CD task.

E. Proxy CD Learning

The proxy CD learning is proposed to guide the LCL to learn features involving the CD task for the encoder. To do this, the most natural way is to perform the CD task supervised by the CD labels. Thus, as shown in the pretraining stage of Fig. 3, the encoder is also asked to identify change and no change in a CD guider on the top of the upsampled projector, which is supervised by the CD ground truth, when the LCL is working. To achieve this, the CD guider is only implemented by a simple CD head, exists in the pretraining stage, and is soon dropped in the later fine-tuning stage. Thus, the behavior of the CD guider can be called proxy CD learning.

Fig. 6 describes the CD guider in detail. First, a 3×3 convolution is used to fuse the information of the upsampled projections u_1 and u_2. Pixelwise identification of change or no change is conducted by a binary classifier of a 1×1 convolution. As a result of the proxy CD, the classifier outputs a proxy change map $\hat{Y}_P$. Due to the downsampling operation in the encoder, the proxy change map $\hat{Y}_P$ and the upsampled projection u_2 are of the same size. In this context, with the CD ground truth Y_{GT} being downsampled to this size, the pixelwise proxy CD loss L_{PCD} can be computed. In the proposed approach, L_{PCD} is simply implemented as cross-entropy loss function

$$L_{\mathrm{PCD}}\left(\hat{Y}_P, Y_{\mathrm{GT}}\right) = \frac{1}{n_p}\sum_{i,j}\tilde{Y}_{\mathrm{GT}}(i,j)\log\hat{Y}_P(i,j) + \left(1-\tilde{Y}_{\mathrm{GT}}(i,j)\right)\log\left(1-\hat{Y}_P(i,j)\right) \quad (12)$$

where $\tilde{Y}_{\mathrm{GT}}$ is a downsampled version of the CD ground truth Y_{GT}. n_p is the number of pixels of $\tilde{Y}_{\mathrm{GT}}$. i and j are the spatial indexes of $\hat{Y}_P$ and $\tilde{Y}_{\mathrm{GT}}$.

IV. Experiments

All the experiments were implemented in PyTorch [37] on an NVIDIA Tesla P100 graphics processing unit (GPU). The proposed SCPFCD framework and the CDContrast pretraining method were evaluated on three public high-resolution remote sensing image CD datasets, i.e., the WHU building dataset [22], the season-varying dataset [38], and the SEmantic Change detectiON Dataset (SECOND) [39].

A. Experimental Settings

1) Datasets:

a) WHU building dataset: The WHU building CD dataset consists of two different temporal remote sensing images of size 32 507 × 15 354, with a spatial resolution of 0.2 m, covering Christchurch, New Zealand. This dataset is only concerned with the changes in buildings. The dataset was divided into training, validation, and test parts according to the ratio of 6:1:3, along the side of 32 507 pixels long. The training part was cropped into patches of size 256 × 256 with an overlapping stride of 128, while the validation and test parts were cropped into patches of size 256 × 256, without overlap. Finally, 17 818, 708, and 2242 bitemporal remote sensing image pairs were obtained for the training set, the validation set, and the test set.

b) Season-varying dataset: The bitemporal remote sensing images of the season-varying dataset were obtained from Google Earth (Digital Globe). The dataset consists of 10 000 bitemporal image pairs of RGB bands for training, 3000 pairs for validation, and 3000 for testing. These images are of size 256 × 256, with a spatial resolution of 0.1–3 m. The mixed spatial resolutions allow various changes to exist in the season-varying dataset, including roads, cars, and large construction structures. Most importantly, due to seasonal changes, the first and second temporal images can be considerably different, such as a sunny day and snow. Therefore, the dataset means that it is a challenge to precisely identify the changes via a CD model.

c) SECOND dataset: The SECOND dataset simultaneously provides CD labels and land-cover labels of six classes, which are nonvegetated ground surface, tree, low vegetation, water, buildings, and playgrounds. Currently, the SECOND dataset provides 2968 bitemporal remote sensing image pairs of size 512 × 512. The images are RGB, with a spatial resolution of 0.5–3 m. The 2968 image pairs of the released training dataset are divided into 2000, 468, and 500 pairs for the training set, the validation set, and the test set, respectively.

2) Evaluation Metrics: In this article, following the representative research [4], [7], [10], [40], the CD performance is evaluated by four commonly used metrics, which are the F_1 score, IoU, Kappa, precision (Pre), and recall (Rec).

3) Benchmarks: The proposed SCPFCD framework and the CDContrast pretraining method were applied to four Siamese networks with different encoders and decoders, namely, FC-Siam-conc (FC-SC) [7], FC-Siam-diff (FC-DIFF) [7], IFN [10], and SNUNet with a width of 24 (SNUNet/24) [9]. Following the original implementation of the afore-mentioned methods, FC-SC and FC-DIFF used the same encoder of four cascaded convolution-ReLu-BN blocks (4-conv), IFN used the VGGNet-16 encoder, and SNUNet used the five residual unit structure (5-Res) encoder.

Furthermore, a ResNet-18 encoder was used for FC-SC and IFN to verify the adaptability of the proposed framework with a commonly used residual encoder and also to compare with

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the encoder that was unsupervisedly pretrained on 1M multitemporal remote sensing images, referring to seco1M[1] [15].

4) Training and Validation Settings: Because the proposed CDContrast pretraining method is fully supervised, the supervised contrastive pretraining stage and fine-tuning stage share the same training settings, including learning rate, weight decay, optimizer, and the maximum epochs, which is inspired by literature [24]. Meanwhile, in the fine-tuning stage, the training settings of the Siamese networks were identical to that of training Siamese networks initialized by random values or ImageNet pretrained model. Namely, the training settings of the CDContrast pretraining, fine-tuning, and benchmark methods are the same. After pretraining, the last checkpoint of the CDContrast pretraining method was used to initialize the encoder of the Siamese networks [26], [27], [28]. The implementation details for the different networks and different datasets are given as follows.

a) FC-SC, FC-DIFF, and IFN: The encoder in pretraining and the Siamese network in fine-tuning were optimized by SGD with a momentum of 0.9, the learning rate set to 0.01, and a weight decay of 4×10^{-4}. In both pretraining and fine-tuning, for the WHU building dataset, the batch size and total epochs were 4 and 18, while the learning rate was decreased by a factor of 10 at the ninth, 14th, and 16th epochs. For the season-varying dataset, the batch size and total epochs were 4 and 20, and the learning rate was decreased by a factor of 10 at the tenth, 15th, and 18th epochs. For the SECOND dataset, the batch size and total epochs were 2 and 30. The learning rate was decreased by a factor of 10 at the 20th, 26th, and 28th epochs.

b) SNUNet/24: To be consistent with the original paper of SNUNet,[2] with the initial learning rate set to 0.001 and decayed by 0.5 every eight epochs and a weight decay of 1×10^{-2}, the encoder in pretraining and the Siamese network in fine-tuning were optimized by the Adam optimizer [41], while the batch size was 16 for the WHU building and season-varying datasets, and 8 for the SECOND dataset. The epochs are 18, 100, and 30 for the WHU building, season-varying, and SECOND datasets. In particular, the SNUNet was trained by the sum of dice loss and focal loss in fine-tuning as in the original paper.

Each checkpoint of the Siamese network was validated on the validation set of the dataset. Next, the checkpoint of the best CD performance on the validation set was further tested on the test set, the performance of which was recorded as the final evaluation result.

B. Experiment 1: The WHU Building Dataset

The visualization results of the predicted change map for the WHU building dataset are shown in Fig. 7, where rows 1–3 show that using a CDContrast pretrained encoder can significantly reduce the false alarms caused by the multitemporal radiation difference, complex textures, and other man-made constructions. Fig. 7 row 4 shows that the parameter initialization provided by the CDContrast pretraining method can help the Siamese network improve the recall ability for the situation of a complex land-cover distribution in the WHU building dataset.

TABLE I
QUANTITATIVE RESULTS OBTAINED ON THE WHU BUILDING DATASET

ID	Algorithm	Init	F_1	IoU	Kappa	Pre	Rec
1	FC-DIFF	from scratch	76.88	62.45	76.18	70.69	84.27
2		**CDContrast**	**84.62**	**73.33**	**84.18**	**82.05**	**87.35**
3	FC-SC	from scratch	72.52	56.89	71.63	61.97	**87.40**
4		**CDContrast**	**83.64**	**71.88**	**83.18**	**82.93**	84.36
5	SNUNet/24	from scratch	72.35	56.68	71.44	61.07	**88.74**
6		**CDContrast**	**80.27**	**67.04**	**79.68**	**74.54**	86.96
7	FC-SC (res18)	ImageNet	84.52	73.18	84.07	81.25	**88.06**
8		**ImageNet +CDContrast**	**85.04**	**73.97**	**84.62**	**85.46**	84.62
9	IFN (res18)	seco1M	83.98	72.38	84.51	78.86	**89.81**
10		ImageNet	84.39	72.28	83.95	83.62	85.17
11		**ImageNet +CDContrast**	**85.17**	**73.49**	**84.75**	**84.28**	86.08
12	IFN (vgg16)	ImageNet	87.49	77.77	87.17	**91.88**	83.51
13		**ImageNet +CDContrast**	**88.37**	**79.16**	**88.05**	90.18	**86.63**

The quantitative experimental results of the proposed framework with the corresponding baseline methods for the WHU building dataset are shown in Table I. With the encoder parameters initialized randomly, FC-DIFF, FC-SC, and SNUNet/24 have high false alarm rates due to the large reflectance variability among the land covers of the WHU building dataset of 0.2-m resolution. From experiments 1–6 of Table I, we can see that CDContrast with random initialization of the encoders improves the CD performance by 7.73%–11.12% in F_1 score, 11.36%–20.96% in the precision score, and 10.36%–14.99% in IoU score compared with the models without the CDContrast pretraining method for their encoders.

For FC-SC (Res18), IFN (Res18), and IFN (VGG16), ImageNet model initialization of the encoder parameters was considered when using the SCPFCD framework and the CDContrast pretraining method, as shown in experiments 7, 8, and 10–13 in Table I. On the shoulders of the powerful ImageNet pretrained model, the CDContrast pretraining method further strengthens the encoder, resulting in an F_1 score improvement of 0.52%–0.88% and an IoU score improvement of 0.78%–1.39%. These experimental results demonstrate that the proposed SCPFCD framework and the CDContrast pretraining method are also effective when an ImageNet pretrained model is used. Notably, as shown in experiments 9–11 of Table I, the F_1 score achieved by IFN (Res18), initialized by seco1M, is even 0.41% worse than that of the ImageNet initialization, indicating that the well-pretrained model on low- and medium-resolution images may not work well for the CD task on high-resolution images. Meanwhile, using ImageNet + CDContrast parameter initialization results in a 1.19% improvement in $F1$ and 1.11% improvement in IoU scores over seco1M, without collecting extra data and huge sources. These experimental results confirm the effectiveness of the proposed SCPFCD framework and the proposed CDContrast pretraining method.

[1]https://github.com/ElementAI/seasonal-contrast
[2]https://github.com/likyoo/Siam-NestedUNet

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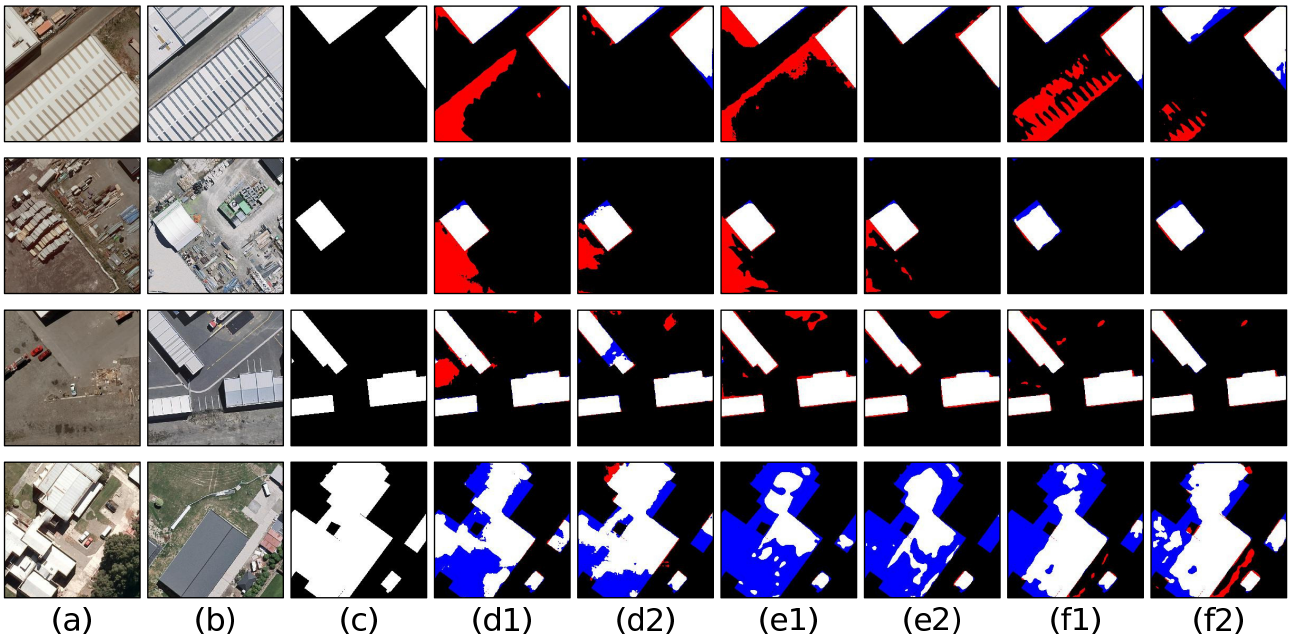


Fig. 7. Visualization results obtained on the WHU building dataset. (a)–(c) First temporal image T_1, the second temporal image T_2, and the CD ground truth, respectively. (d)–(f) Visualization results for SNUNet, FC-SC (Res18), and IFN (VGG16). In (d)–(f), "*1" and "*2" represent the change maps predicted by the specific Siamese network without and with the CDContrast pretraining, respectively (□ TPs, ■ FPs, ■ FNs, and ■ TNs).

C. Experiment 2: The Season-Varying Dataset

Fig. 8 shows the visualization results obtained on the season-varying dataset, which demonstrates the superiority of the proposed framework when extracting delicate changes from high-resolution bitemporal data with a lot of changes of no interest, such as snow.

The quantitative experimental results for the season-varying dataset are listed in Table II. When equipped with the encoder pretrained by CDContrast with the random initialization, the performances of FC-DIFF and FC-SC show improvements in the F_1 score of 3.57% and 2.38% and in the IoU score of 7.50% and 9.04%, respectively. Even though the F_1 and IoU scores of SNUNet/24 have reached 94.44% and 89.47%, which is nearly saturated, the CDContrast parameter initialization can further improve its F_1 and IoU scores to 95.11% and 90.67%. When the CDContrast pretraining method is initialized by an ImageNet pretrained model, as shown in experiments 7, 8, and 10–13 in Table II, the encoder pretrained by CDContrast shows improvements of 0.57%–1.11% and 0.73%–2.09% in terms of F_1 and IoU scores, respectively. Table II experiments 9–11 further verify that the CDContrast parameter initialization surpasses the existing CL method for multitemporal images, seco1M, by 1.25% in the F_1 score and 1.93% in the IoU score. The experimental results obtained on the season-varying dataset further demonstrate the effectiveness of the proposed framework.

TABLE II
QUANTITATIVE RESULTS OBTAINED ON THE SEASON-VARYING DATASET

ID	Algorithm	Init	F_1	IoU	Kappa	Pre	Rec
1	FC-DIFF	from scratch	79.99	70.10	76.87	73.19	88.19
2		**CDContrast**	**83.57**	**77.60**	**81.12**	**79.24**	**88.41**
3	FC-SC	from scratch	79.95	68.38	76.91	75.01	85.57
4		**CDContrast**	**82.33**	**77.42**	**79.63**	**76.62**	**88.95**
5	SNUNet/24	from scratch	94.44	89.47	93.70	95.15	93.74
6		**CDContrast**	**95.11**	**90.67**	**94.46**	**95.65**	**94.56**
7	FC-SC (res18)	ImageNet	85.27	74.32	82.90	76.01	**97.10**
8		**ImageNet +CDContrast**	**85.84**	**75.19**	**83.64**	**78.91**	94.10
9	IFN (res18)	seco1M	85.15	74.13	82.99	**83.00**	87.41
10		ImageNet	85.62	74.86	83.32	76.67	**96.95**
11		**ImageNet +CDContrast**	**86.40**	**76.06**	**84.30**	79.78	94.22
12	IFN (vgg16)	ImageNet	90.64	82.70	89.32	90.51	90.76
13		**ImageNet +CDContrast**	**91.75**	**84.79**	**90.58**	**91.37**	**92.12**

D. Experiment 3: The SECOND Dataset

1) Visualization and Quantitative Experimental Results: Fig. 9 shows the visualization results achieved on the SECOND dataset. The middle regions of the bitemporal images in Fig. 9 rows 1 and 2 are "nonvegetated ground surface to low vegetation" and "nonvegetated ground surface to buildings," respectively, leading to many missed changes due to the similar spectral characteristics in the high-resolution images. As shown in rows 3 and 4 in Fig. 9, the extreme radiation difference and registration error cause varying degrees of false alarms. However, these errors can be partially alleviated when the CDContrast pretrained models are used.

The quantitative experimental results for the SECOND dataset are listed in Table III. Experiments 1–6 show that the proposed framework improves the F_1 score and the IoU score of the downstream CD task by 0.97%–2.73% and 1.08%–2.99%, respectively, for the CDContrast pretraining with random initialization. For the CDContrast pretraining with ImageNet model initialization, the improvements in the F_1 and IoU scores are about 0.87%–1.43% and 0.99%–1.24% in experiments 7, 8, and 10–13. Table III experiments 9–11

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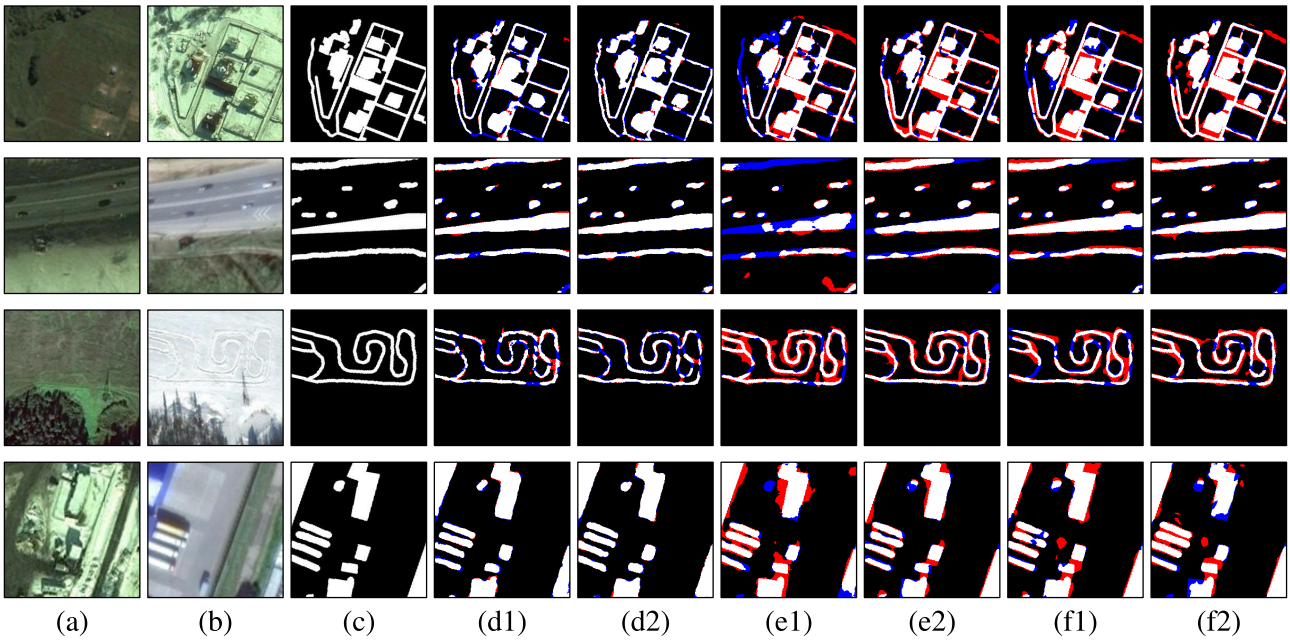


Fig. 8. Visualization results obtained on the season-varying dataset. (a)–(c) First temporal image T_1, the second temporal image T_2, and the CD ground truth. (d)–(f) Visualization results for SNUNet, FC-SC (Res18), and IFN (VGG16). In (d)–(f), "*1" and "*2" represent the change maps predicted by the specific Siamese network without and with the CDContrast pretraining, respectively (□ TPs, ■ FPs, ■ FNs, and ■ TNs).

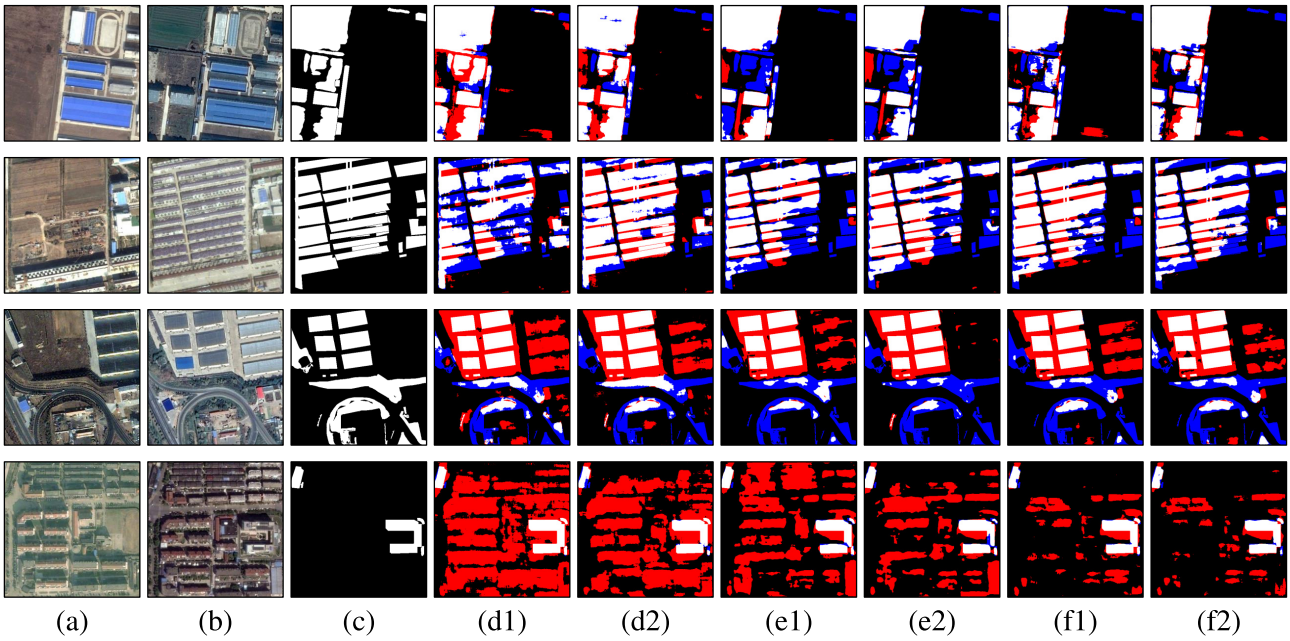


Fig. 9. Visualization results obtained on the SECOND dataset. (a)–(c) First temporal image T_1, the second temporal image T_2, and the CD ground truth. (d)–(f) Visualization results for SNUNet, FC-SC (Res18), and IFN (VGG16). In (d)–(f), "*1" and "*2" represent the change maps predicted by the specific Siamese network without and with the CDContrast pretraining, respectively (□ TPs, ■ FPs, ■ FNs, and ■ TNs).

show that the F_1 and IoU scores of the ImageNet + CDContrast parameter initialization are 1.25% and 0.77% higher than that of seco1M, which further verifies the superiority of the CDContrast pretraining method compared with the seco1M initialization. The reason why the proposed framework works so well is revealed in the following.

2) Impact of the CDContrast Parameter Initialization on the Encoder and the Decoder of the Downstream CD Task: Inspired by Wang et al. [42] and Wang et al. [43], feature statistics and feature visualization were used to analyze the features and study the impacts of the different parameter initialization methods. To this end, the features of FC-SC and FC-SC (Res18) with different initializations, i.e., corresponding to Table III experiments 3, 4, 7, and 8, were further analyzed and compared.

The encoder features, which were output by the last layer of the encoders, were analyzed to explore the spatial relationships among the land cover of each temporal remote sensing image. Due to the downsampling of many max-pooling layers, the encoder features were of low spatial resolution and only roughly reflected the representation ability of different encoders for the land covers. In this context, as shown in

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TABLE III
QUANTITATIVE RESULTS OBTAINED ON THE SECOND DATASET

ID	Algorithm	Init	F_1	IoU	Kappa	Pre	Rec
1	FC-DIFF	from scratch	63.26	46.26	55.77	71.88	56.49
2		**CDContrast**	**66.00**	**49.25**	**58.91**	**73.61**	**59.82**
3	FC-SC	from scratch	63.06	46.05	55.64	72.53	55.78
4		**CDContrast**	**65.18**	**48.34**	**58.12**	**74.37**	**58.00**
5	SNUNet/24	from scratch	65.43	48.62	57.79	**69.57**	61.75
6		**CDContrast**	**66.40**	**49.70**	**58.80**	69.22	**63.81**
7	FC-SC (res18)	ImageNet	67.16	50.56	60.58	**77.35**	59.35
8		**ImageNet +CDContrast**	**68.03**	**51.55**	**61.28**	75.09	**62.18**
9	IFN (res18)	seco1M	66.60	49.92	59.78	75.60	**59.52**
10		ImageNet	66.38	49.68	60.02	**80.25**	56.60
11		**ImageNet +CDContrast**	**67.28**	**50.69**	**60.72**	77.48	59.46
12	IFN (vgg16)	ImageNet	67.36	50.67	60.94	**78.87**	58.78
13		**ImageNet +CDContrast**	**68.79**	**51.91**	**62.44**	78.28	**61.36**

columns 4–6 in Table IV, the encoder features of FC-SC and FC-SC (Res18) were used to calculate the mean intraclass distance $D_{\mathrm{LC}}^{\mathrm{intra}}$ and the mean interclass distance $D_{\mathrm{LC}}^{\mathrm{inter}}$ of the land cover by the following equation:

$$D_{\mathrm{LC}}^{\mathrm{intra}} = \frac{1}{N}\sum_{i=1}^{N} d_i^{\mathrm{intra}}, \quad D_{\mathrm{LC}}^{\mathrm{inter}} = \frac{1}{N}\sum_{i=1}^{N} d_i^{\mathrm{inter}}. \tag{13}$$

Here, the intraclass distance d_i^{intra} of the ith class was calculated by the mean distance between the intraclass features and the feature center $\mathbf{c}_i$ of the ith class, while the interclass distance d_i^{inter} of the ith class was defined by the distance between the centers of the different classes

$$d_i^{\mathrm{intra}} = \frac{1}{n_i}\sum_{t=1}^{n_i} ||\mathbf{f}_t - \mathbf{c}_i||_2^2 \tag{14}$$

$$d_i^{\mathrm{inter}} = \frac{1}{N-1}\sum_{i=1, i\neq j}^{N} \left|\left|\mathbf{c}_i - \mathbf{c}_j\right|\right|_2^2 \tag{15}$$

$$\mathbf{c}_i = \frac{1}{n_i}\sum_{t=1}^{n_i} \mathbf{f}_t \tag{16}$$

where n_i is the number of pixels of the ith class, $\mathbf{f}_t$ is the tth pixel of the ith class in the test set, and N is the number of classes. In addition, inspired by Wang et al. [42] and Zhao et al. [44], the interclass distance standard deviation (std) $S_{\mathrm{LC}}^{\mathrm{inter}}$ was adopted to measure the feature distribution uniformity of the different classes because better uniformity means better discrimination of the different-class features for classification [42]

$$S_{\mathrm{LC}}^{\mathrm{inter}} = \left[\frac{1}{N-1}\sum_{i=1}^{N}\left(D_{\mathrm{LC}}^{\mathrm{inter}} - d_i^{\mathrm{inter}}\right)^2\right]^{0.5}. \tag{17}$$

A smaller value of S^{inter} means better interclass uniformity.

For the decoder feature, it directly reflects the CD ability because the decoder extracts change information from the spatial features of bitemporal images. As shown in columns 7–9 in Table IV, the decoder features were used to calculate the interclass distance $D_{\mathrm{CD}}^{\mathrm{inter}}$ between "no change" and "change," the "no change" intraclass distance $d_{\mathrm{unc}}^{\mathrm{intra}}$, and the "change" intraclass distance $d_{\mathrm{ch}}^{\mathrm{intra}}$ by (13)–(15), respectively.

Focusing on the CD ability from the perspective of the decoder features, it can be easily found from columns 7 to 10 of rows 2 and 3 in Table IV that FC-SC with encoder pretrained by CDContrast increases the interclass distance $D_{\mathrm{CD}}^{\mathrm{inter}}$ between "no change" and "change" by 10.99%, leading to a CD performance improvement of 2.12% in terms of the F_1 score. In contrast, as shown in columns 7–10 of rows 4 and 5 in Table IV, FC-SC (Res18) with encoder pretrained by ImageNet + CDContrast reduces the intraclass distances of both "no change" and "change," bringing a CD F_1 score improvement of 0.87%. In addition, columns 8–10 in Table IV also show that the interclass distance between change and no change is larger than the intraclass distance of change but is less than the sum of intraclass distances of the change and no change. This means that the two clusters of change and no change are separated but partially mixed, which accounts for the unsatisfying CD performance of F_1 scores that are only more than 60%.

When it comes to the spatial relationships among the land cover, columns 4–6 of rows 2–5 in Table IV show that both the CDContrast and ImageNet + CDContrast methods considerably reduce the mean intraclass distance $D_{\mathrm{LC}}^{\mathrm{intra}}$ and increase the interclass distance std S^{inter} of the land cover compared with random initialization and ImageNet pretrained model initialization. Here, due to the low resolution of the encoder features, although Table IV also shows larger intraclass distances than interclass distances of the land covers, it reflects the main characteristics of the land-cover clusters.

The experimental results consistently indicate that the CDContrast pretraining method reduces the land-cover intraclass distance and increases the large land-cover interclass distance, so as to improve the Siamese networks for the CD task, in terms of the spatial relationships among the land cover. With regard to the CD ability, the random values and the ImageNet pretrained model show different requirements for decoder feature characteristics when performing the CD task, but the proposed CDContrast pretraining method is compatible with them both and provides a better pretrained encoder for the downstream CD task.

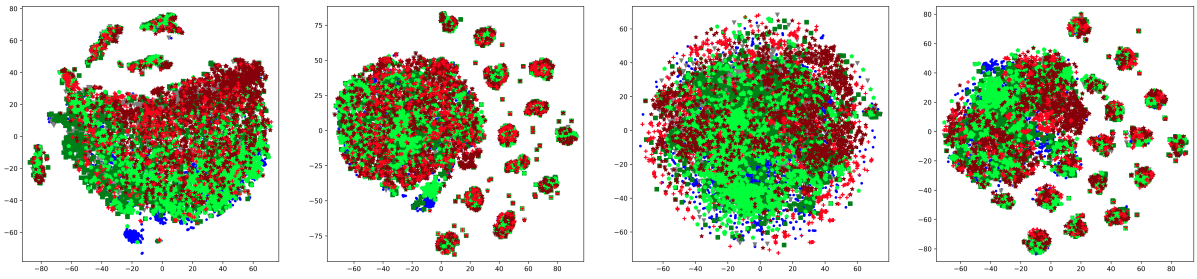
3) Key Indicators for Encoder Initialization: Without fine-tuning, the output features of the encoders with the four initialization methods were used to compute the mean intraclass distance $D_{\mathrm{LC}}^{\mathrm{intra}}$, the mean interclass distance $D_{\mathrm{LC}}^{\mathrm{inter}}$, and the interclass distance std $S_{\mathrm{LC}}^{\mathrm{inter}}$ of the land cover, to find out which statistics influence the downstream task most, as shown in columns 2–4 in Table V. In addition, a UNet model [25], which was initialized with the four methods, was fine-tuned to perform the land-cover classification task, supervised by the land-cover labels of the SECOND dataset. The mean Intersection over Union (mIoU) was used to evaluate the land-cover classification performance, as shown in column 5 (LC mIoU) in Table V.

In rows 2 and 3 Table V, the CDContrast pretrained model improves the mIoU of the land-cover classification by 6.71% and improves the F_1 score of the CD by 2.12% compared

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TABLE IV
FEATURE ANALYSIS FOR THE SIAMESE NETWORKS

CD algorithm	Encoder	Init	Encoder feature			Decoder feature			CD F_1
			D_{LC}^{intra}	D_{LC}^{inter}	S_{LC}^{inter}	d_{unc}^{intra}	d_{ch}^{intra}	D_{CD}^{inter}	
FC-SC	4-conv	from scratch	11.5067	2.1335	1.1664	1.2578	2.1117	1.8429	63.06
		CDContrast	1.1806	0.2890	0.1519	1.3586	2.3689	2.0454	65.18
FC-SC	ResNet-18	ImageNet	2.0005	0.2739	0.0947	1.4857	1.8122	1.6855	67.16
		ImageNet+CDContrast	1.8084	0.2311	0.0822	1.4340	1.5849	1.5289	68.03



(a) (b) (c) (d)

Fig. 10. Encoder feature t-SNE visualizations of the four parameter initialization methods (water, nonvegetated ground surface, low vegetation, tree, buildings, and playgrounds). (a) From scratch. (b) Proposed CDContrast pretraining method. (c) ImageNet. (d) ImageNet + the proposed CDContrast pretraining method.

TABLE V
ENCODER FEATURE ANALYSIS FOR THE FOUR PARAMETER INITIALIZATION METHODS

Init	D_{LC}^{intra}	D_{LC}^{inter}	S_{LC}^{inter}	LC mIoU	CD F_1
4-conv (from scratch)	1.9414	0.4710	0.5113	40.86	63.06
4-conv (CDContrast)	6.3239	0.1016	0.0626	47.57	65.18
ResNet-18 (ImageNet)	7.1603	0.4479	0.2216	54.34	67.16
ResNet-18 (ImageNet+ CDContrast)	10.5549	0.3124	0.1809	55.04	68.03

with random value initialization. In rows 4 and 5 Table V, the ImageNet + CDContrast pretrained model improves the mIoU of the land-cover classification by 0.70% and improves the F_1 score of the CD by 0.87% compared with the ImageNet pretrained model.

Comparing row 2 with row 3 in Table V and comparing row 4 with row 5 in Table V, it can be observed that the CDContrast pretraining method unexpectedly increases the intraclass distance and reduces the interclass distance but significantly reduces the interclass distance std, meaning an improvement of the interclass uniformity. From rows 2 and 5 in Table V, it can be further found that "ResNet-18 (ImageNet + CDContrast)" outputs features with a larger land-cover interclass distance than "4-conv (CDContrast)." These experimental results indicate that the interclass distance and the interclass uniformity of the features may be the key indicators for a pretrained encoder and play an important part in the performance of the downstream CD task.

To enhance the understanding of the different parameter initialization methods, as shown in Fig. 10, the land-cover features extracted by the encoders with the four different initialization methods in Table V are visualized in 2-D space using t-SNE [45]. The different colors denote the feature points of the different land-cover classes in the SECOND dataset. Comparing Fig. 10(a) with (b) and comparing Fig. 10(c) with (d), it can be seen that the proposed CDContrast pretraining method makes the features locally aggregate into many clusters, which accounts for the intraclass distance increase and the interclass distance reduction of the land cover in Table V.

V. ABLATION STUDY

Using the FC-SC and IFN (Res18) networks, which are equipped with totally different encoders and decoders, an ablation study was conducted to evaluate the influence of each part, the maximum number of training epochs, and the number of training samples on the downstream CD performance.

A. Effectiveness of Each Part

As shown in Table VI, we varied the encoder initialization methods (column 3) and the upsampling scale factor (column 4) of the two Siamese networks (column 2) and compare the total CD performances by the F_1 scores (column 5), which comprehensively considers the precision and recall scores. The loss curves during the pretraining stage and validation curves during the fine-tuning stage are shown in Fig. 11, where 80 000 iterations denote the maximum pretraining and fine-tuning epochs of 18 on the WHU building dataset.

1) FC-SC: Experiments 1–7 were conducted using the FC-SC network with the 4-conv encoder, which randomly initializes its parameters before the pretraining stage. Without any pretraining in experiment 1, FC-SC achieves an F_1 score of 72.52%. In experiment 2, the classic contrastive loss [40] of metric learning was used to pretrain the encoder. This loss constructs similar sample pairs to those of LCL, but its performance is much worse than the model from scratch. This is because the metric learning losses, such as contrastive

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TABLE VI
ABLATION STUDY WITH THE WHU BUILDING DATASET. THE BEST F_1 SCORES ARE MARKED IN BOLD

ID	Algorithm	Init	Upsample	F_1
1	FC-SC	from scratch	-	72.52
2		metric learning	1	56.19
3		LCL	1	77.02
4		LCL	2	80.52
5		CDContrast (LCL+P.CD.L)	1	82.31
6		**CDContrast (LCL+P.CD.L)**	**2**	**83.64**
7		CDContrast (LCL+P.CD.L)	4	80.84
8	IFN (Res18)	from scratch	-	74.50
9		metric learning	1	58.73
10		LCL	1	79.55
11		LCL	2	80.01
12		CDContrast (LCL+P.CD.L)	1	80.35
13		CDContrast (LCL+P.CD.L)	2	81.82
14		CDContrast (LCL+P.CD.L)	4	81.43
15		ImageNet	-	84.39
16		ImageNet +LCL	2	75.49
17		**ImageNet +CDContrast (LCL+P.CD.L)**	**2**	**85.17**

P.CD.L refers to the proxy CD learning.

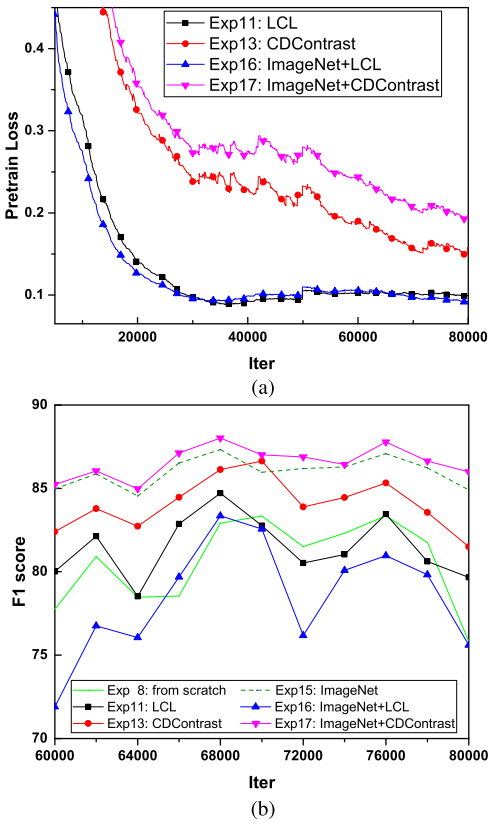


Fig. 11. Pretraining and fine-tuning curve. (a) Pretraining loss of the LCL and the CDContrast pretraining method on the training set of the WHU building dataset, using the ResNet-18 encoder, for 80 000 iterations, i.e., 18 epochs. (b) F_1 score curve of the IFN (Res18) network on the validation set of the WHU building dataset from 60 000 to 80 000 iterations (14–18 epochs).

loss and triplet loss [46], only use one negative sample in a sample pair, significantly limiting the power of the learned features [30].

Meanwhile, for the proposed LCL that is under the CL mechanism of the asymmetric networks, its pretrained encoder improves the F_1 score by 4.50% from experiments 1 and 3. Comparing experiments 3 and 5, it can be found that the CD guider introduced by the proxy CD learning improves the LCL, with an improvement in F_1 score of 5.29%. Experiments 3–7 demonstrate that the upsampling projector is also effective. When the upsampling scale factor is set to 2, the F_1 score improvement reaches the maximum of 2.98% for experiments 4 and 6. It can also be observed from experiments 4 and 5 that the F_1 score even reduces when the upsampling factor is set to 4. Here, we conjecture that the ability of the upsampling projector to restore the spatial resolution of the feature map is limited, as it fails to fully use the Y_{GT} of a high spatial resolution and suffers from noise in the CDContrast pretraining. We did not try a scale factor larger than 4 because the GPU could not cope with the exponential memory usage of the upsampled projection.

These experimental results demonstrate the effectiveness of the proposed SCPFCD framework and the CDContrast pretraining method for the CD task.

2) IFN (Res18): Experiments 8–17 were conducted using the IFN (Res18) network with the ResNet-18 encoder. The conclusion obtained from experiments 8 to 14 is similar to that for experiments 1–7 when the ResNet-18 encoder used random initialization in the pretraining stage. In experiments 15–17, we explored the impact of the SCPFCD framework and the CDContrast pretraining method when an ImageNet pretrained model was used for the parameter initialization. In this context, the ResNet-18 encoder with ImageNet pretrained model initialization was first pretrained by the LCL or CDContrast in the pretraining stage and was then trained with a complete CD network to perform the CD task in the fine-tuning stage.

However, experiments 15 and 16 show that the LCL undermines the feature extraction ability of the ImageNet pretrained model, decreasing the F_1 score by 7.73%. Fig. 11(b) also shows that, nearly, the whole validation curve of the LCL with ImageNet initialization (Exp16: ImageNet + LCL) is below the curve of ImageNet initialization alone (Exp15: ImageNet). From Fig. 11(a), it can also be seen that the loss value of LCL with ImageNet initialization (Exp16: ImageNet + LCL) is even lower than that of LCL (Exp11: LCL) alone at the end of the training. This indicates that the strong ImageNet initialization possibly misleads the encoder to focus on the single-temporal remote sensing image too much, and it neglects to learn features for the CD task.

However, when the encoder is pretrained by the proposed CDContrast method, where the proposed proxy CD learning is introduced into the LCL, things change. As shown in Fig. 11(a), the pretraining loss of CDContrast with ImageNet initialization (Exp17: ImageNet + CDContrast) considerably increases when compared with LCL with ImageNet initialization (Exp16: ImageNet + LCL). Meanwhile, as can be seen in Fig. 11(b), the validation curve of CDContrast with ImageNet

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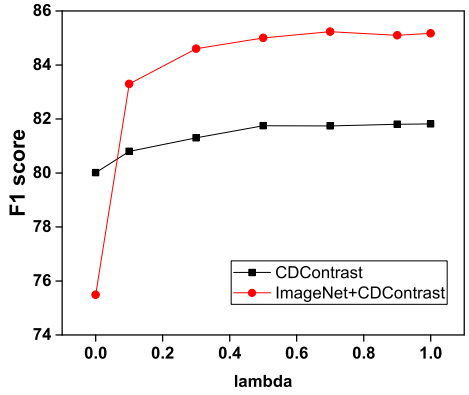


Fig. 12. F_1 score curve of the IFN (Res18) network on the testing set of the WHU building dataset with different λ's in the pretext task loss L_{pt} of the CDContrast pretraining method. λ controls the ratio of the proxy CD loss L_{PCD}.

initialization (Exp17: ImageNet + CDContrast) is consistently above the curve of LCL with ImageNet initialization (Exp16: ImageNet + LCL). These experimental results demonstrate the significance of the proxy CD learning in avoiding a local optimum of LCL, especially when the ImageNet pretrained model is used.

To explore the influence of the different ratios of the proposed proxy CD learning in the pretext task loss L_{pt}, its weight hyperparameter λ was varied from 0 to 1 on the WHU building dataset with the IFN (Res18) method. As shown in Fig. 12, the improvements of F_1 score brought by the proposed proxy CD learning nearly reach the maximums when $\lambda > 0.5$.

B. Maximum Number of Training Epochs

A large number of training epochs for unsupervised CL pretraining methods [27], [28] inspired us to verify the influence of the training epochs on the proposed CDContrast pretraining method. In the experiments section, for the WHU building dataset, the SCPFCD framework pretrained the encoders for 18 epochs and fine-tuned the Siamese networks for 18 epochs. As shown in Fig. 13, the CDContrast pretraining epoch number varied from 5 to 200, while the fine-tuning epoch number was still 20. Compared with FC-SC with the random initialization, FC-SC with CDContrast pretraining initialization consistently achieves more than 8% higher F_1 scores. Compared with IFN (Res18) with ImageNet pretrained model initialization, IFN (Res18) with CDContrast based on ImageNet pretrained model initialization achieves around 1% higher F_1 scores when the number of pretraining epoch numbers is bigger than 5 and less than 100. The experimental results show that the fully supervised CDContrast suffers too many pretraining epochs, which is different from those unsupervised CL pretraining methods [27], [28].

Therefore, the maximum epoch number of CDContrast pretraining is set to the epoch number of training an end-to-end CD network initialized with random values or ImageNet pretrained model, which is enough for the convergence.

C. Number of Training Samples

To verify the influence of the number of training samples, FC-SC and INF (Res18), which were initialized with and

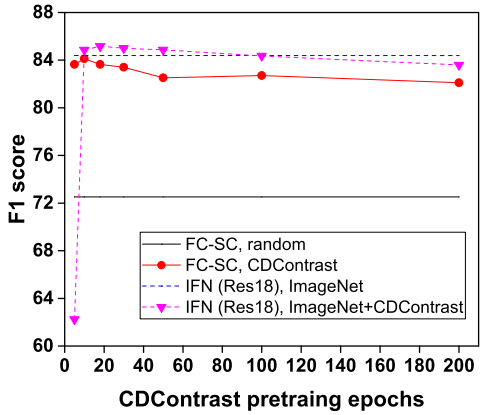


Fig. 13. Verification of different pretraining epochs on the CD networks. The encoders were pretrained by CDContrast with 5, 10, 20, 30, 100, and 200 epochs. Initialized the pretrained encoders, the Siamese networks were then fine-tuned for 18 epochs.

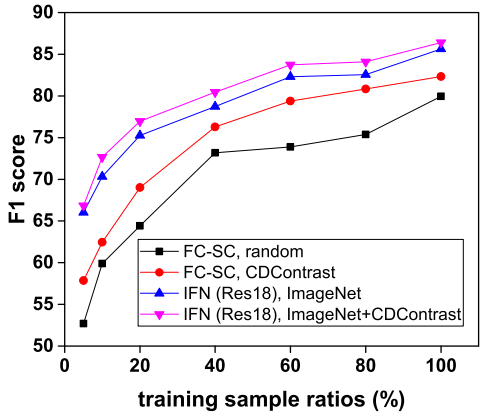


Fig. 14. Verification of the influence of different training samples on the proposed method. Here, 5%, 10%, 20%, 40%, 60%, and 80% of the season-varying datasets training set are used to train the Siamese network with and without the proposed CDContrast pretraining method.

without the proposed CDContrast pretraining method, respectively, were trained on 5%, 10%, 20%, 40%, 60%, and 80% of the season-varying datasets training set of 10 000 bitemporal image pairs. As shown in Fig. 14, the F_1 scores of all the methods increase with the number of the training samples increasing, which indicates that the proposed SCPFCD framework is limited by the number of labeled training samples, in line with the perception of general supervised learning. Even so, Fig. 14 also shows that the Siamese networks initialized by the CDContrast can always achieve better CD performance than the Siamese networks initialized with random values and ImageNet pretrained model.

D. Space and Time Complexities

Table VII shows that using CDContrast pretraining only introduces an additional 0.32M and 2.97M parameters for FC-SC and IFN (Res18) compared with training the two Siamese networks directly. In particular, as shown in Fig. 3, the encoder, the upsampling projector, and the predictor are weight-sharing between "contrasting T_2 with T_1" and "contrasting T_1 with T_2" in the proposed CDContrast pretraining method. The momentum encoder and the momentum upsampling projector are parameterless because they are obtained by a moving average from the encoder and the upsampling

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TABLE VII
SPACE AND TIME COMPLEXITIES FOR THE FC-SC AND THE IFN (RES18) ON THE WHU BUILDING DATASET

Method	Parameter/M	Training time/hours
FC-SC	1.46	1.78
FC-SC +CDContrast	1.78	3.33
IFN (Res18)	16.66	7.11
IFN (Res18) +CDContrast	19.63	9.24

projector. Therefore, for a Siamese network with the CDContrast pretraining method, the additional parameters only come from the upsampling projector, the predictor, and the CD guider, which include two, two, and two convolution operators, respectively. The specific additional parameters of these modules are determined by the number of output channels of the encoder features.

Table VII also shows that using the CDContrast pretraining method only costs about an extra 2 h for the pretraining. Although the images need to pass through both the encoder and the momentum encoder, the momentum encoder works without computing gradients [28], [29]. Meanwhile, the upsampling projector, the predictor, and the CD guider are only several convolutions. Thus, the CDContrast pretraining method does not cost too much time before training a Siamese network for the CD task. In addition, the encoder pretrained by the CDContrast pretraining method can be applied to any Siamese network that uses a specific encoder. For example, a pretrained ResNet-18 encoder was used for both IFN and FC-SC in Tables I–III. Thus, there is no need to pretrain the encoder for all the different Siamese networks.

VI. CONCLUSION

In this article, the SCPFCD framework, which consists of a supervised contrastive pretraining stage and a fine-tuning stage, has been proposed to train a Siamese network based on a pretrained encoder for CD. In the supervised contrastive pretraining stage, an encoder is pretrained by the proposed CDContrast pretraining method, which includes the LCL based on supervised CL and proxy CD learning. Using CD labels, the LCL effectively uses bitemporal pixel pairs to construct positive samples for CL to learn the spatial relationships of the land cover. Meanwhile, the LCL is performed using upsampled projections to capture the details of the land covers, alleviating the information loss of downsampled labels. The proposed proxy CD learning, which performs the proxy CD task on the top of the upsampled projector, works together with the LCL, so as to guarantee the feature quality for the downstream CD task. The feature analysis reveals that the CDContrast pretraining method improves the CD ability by increasing the interclass distance and interclass uniformity of the land cover for the encoder initialization. The quantitative experimental results confirmed the superiority of the proposed SCPFCD framework. However, a limitation lies in the paradigm of the LCL, which still cannot fully use the information of the CD labels to construct the sample pairs for the CL. In the future, we will further explore possible effective pretext tasks, such as reconstructing a masked image [47], for the supervised contrastive pretraining, to use CD labels to learn features for bitemporal images.

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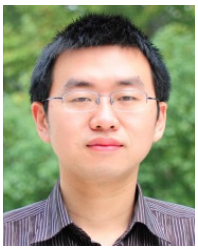
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